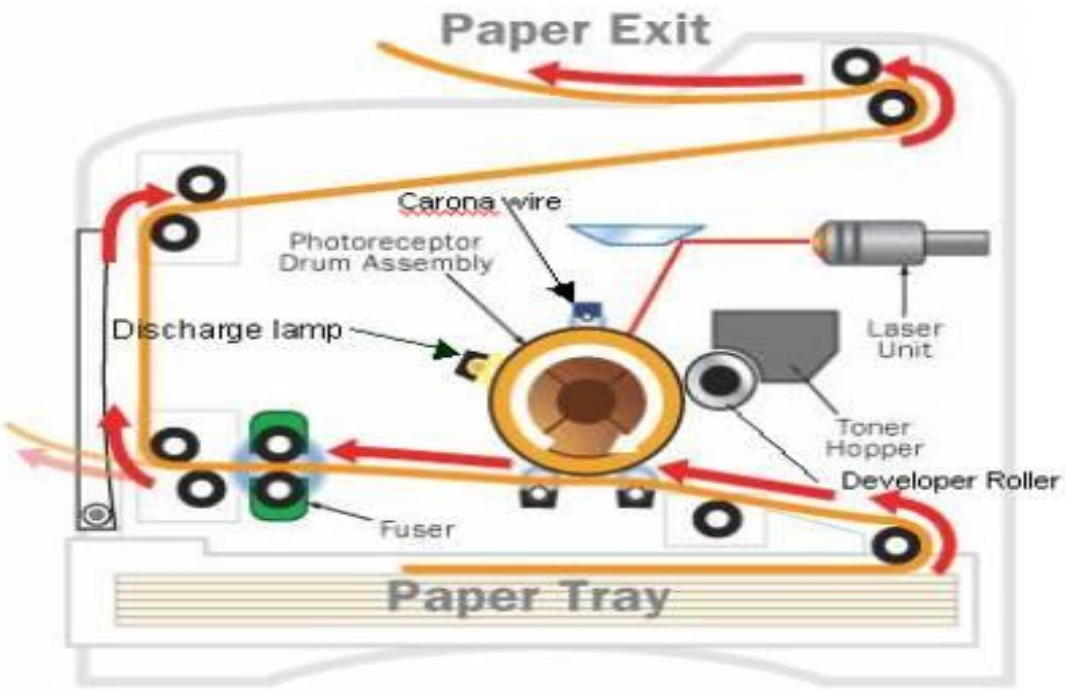
Printer & Scanner Basics

Introduction to Printers

- Printers are commonly used output devices that produce a hard copy of document stored in electronic form, i.e they put information from computer on to paper.
- There are various kinds of printers available today like Impact printers, Bubble-jet printers, Laser printers, Thermal printers etc.
- **Types of printers**
  - Impact
  - Solid-Ink Printers
  - Thermal Printers
  - Inkjet Printers
  - Laser printers

Laser Printer



Laser printer components and their functionality

Introduction to Scanners

- Scanner is a device that scans images, printed text, and handwriting etc and converts it to digital form or image.
- It is so named because the data is converted one line at a time or scanned down the page as the scanning head moves down the page.

Components inside a scanner

- Glass Plate and Cover
- Scanning head
- Stepper Motor

Types

- Flatbed Scanners
- Sheetfed Scanners
- Handheld Scanners

Printer and Scanner Interfaces

- A printer's interface is a combination of hardware and software that allows the printer to communicate with a computer.
- The hardware interface is called port and each printer has atleast one interface. An interface incorporates several components including its communication type as well as the interface software.
- There are eight major communication types

Printer and Scanner Interfaces

1. Serial : Computers send one bit at a time, one after another through serial connection. The communication parameters like parity, baud should be set on both entities before communication takes place.

2. Parallel : Parallel communication is more popular with printers because it is more faster than serial communication. In this type of communication, printer receives eight bits at a time over eight separate wires.

Parallel uses a DB25 connection on the computer side and a oddly shaped 36 pin connection on the printer.

3. USB (Universal Serial Bus) : These can transfer data quickly with a transfer rate up to 12 Mbps and it automatically recognizes new devices.

4. Network : Network, or commonly Ethernet, connections are commonplace on network laser printers, though some other types of printers do employ this type of connection. These printers have a Network Interface Card (NIC) and ROM-based software that allows them to communicate with networks, servers and workstations.

Printer and Scanner Interfaces

5. Infrared : Infrared transmissions are wireless transmissions that use radiation in the infrared range of the electromagnetic spectrum. An Infrared acceptor allows your devices (laptops, PDA's, Cameras, etc) connect to the printer and send print commands via infrared signals.

6. SCSI (Small Computer System Interface) : Laser printer, dye-sublimation printers and few other use SCSI interfaces to PC as there are benefits like daisy chaining wherein more than one device could be on a single SCSI connection and it is easy to implement.

7. IEEE 1394 Firewire : Firewire is a high speed connection typically used for digital video editing or other high bandwidth requirements. This interface currently supports devices with a maximum throughput of 800 Mbps and capable of speeds up to 3.2 Gbps.

8. Wireless : Wireless is the currently popular technology like infrared, Bluetooth, 802.11 etc. The information is transmitted wirelessly through the air using radio waves and is received by the device.

Bluetooth is used to replace the cables between computers and its peripherals and they usually work over small distances of about 10 meters.

Out of these above communication types scanners mostly use USB, Parallel, SCSI, IEEE 1394/FireWire.

Installing And Configuring Printers And Scanners

- Step 1 : Attach the device using a local or network port and connect the power
- Step 2 : Install and Update the Device driver and calibrate the device
- Step 3 : Configure options and default settings
- Step 4 : Print/scan a test page

Printers And Scanners Problems And Troubleshooting

- Dot-Matrix printer problems
  - Print results are too light
  - Printer does not print
  - Characters are incomplete
  - Printout jams inside the printer
  - Stepper motor problems

Printers And Scanners Problems And Troubleshooting

- Laser printer problems
  - Print results too light, white/light columns
  - Ghosting
  - Prints black pages
  - Memory Overflow error
  - Paper jams
  - Repetitive small marks or defects
  - Image Smudging

Scanner Problems and Troubleshooting

- Common problems with scanner are scanners won't turn on, strange noises from scanner, Scanners won't scan etc.
- Tips to troubleshoot the basic scanner related problems are:
  - Ensure that the scanner software is properly installed and all the settings are properly configured
  - Make sure that the power light on the scanner is turned on.
  - If any software related problems are noticed in scanner then uninstall and reinstall the scanner software
  - Try to perform the basic tests that are provided by the manufacturer.
  - Make sure that the cables to the scanner are properly connected.
  - Overheating, dust and humid environment can cause permanent damage to the scanner.
  - Restarting your computer, unplugging cables and verifying any hardware conflicts in the device manager can diagnose and troubleshoot the possible errors.

Tools To Troubleshooting Problems

- Multimeter
- Screw driver
- Cleaning solutions
- Extension Magnets
- Test patterns

References (https://www.tutorialsweb.com/computers/printers-scanners/.htm)

1. https://www.techexams.net
2. https://www.howstuffworks.com
3. https://www.brainbell.com
4. CompTIA A+ complete study guide

Thank you